BUSINESS PROBLEM STATEMENT

Customer Data Analysis

1. Business Problem Statement

The project aims to analyse shopping transactions from different shopping malls to understand customer shopping behaviour. The analysis focuses on shopping patterns by gender and age, product-category preferences, product sales, revenue generation and payment-method usage. The objective is to derive meaningful insights from the data and provide useful suggestions to the company.

2. Project Analysis Tasks

1. How is the shopping distribution according to gender?

2. Which gender did we sell more products to?

3. Which gender generated more revenue?

4. Distribution of purchase categories relative to other columns?

5. How is the shopping distribution according to age?

6. Which age category did we sell more products to?

7. Which age category generated more revenue?

8. Distribution of purchase categories relative to other columns?

9. Does the payment method have a relation with other columns?

10. How is the distribution of the payment method?

11. Visualize the data using Tableau / Power BI and derive insights and give your inputs / suggestions to the company.

3. Dataset Description

This dataset contains shopping information from 10 different shopping malls between 2021 and 2023. It includes information from various age groups and genders to provide a view of shopping habits in Istanbul. The dataset includes invoice numbers, customer IDs, age, gender, payment methods, product categories, quantity, price, order dates and shopping mall locations.

4. Attribute Information

Attribute	Description
invoice_no	A combination of the letter 'I' and a 6-digit integer uniquely assigned to each operation.
customer_id	A combination of the letter 'C' and a 6-digit integer uniquely assigned to each operation.
gender	String variable of the customer's gender.
age	Positive integer variable of the customer's age.
category	String variable of the category of the purchased product.
quantity	The quantities of each product (item) per transaction.
price	Product price per unit in Turkish Liras (TL).
payment_method	String variable of the payment method (cash, credit card or debit card) used for the transaction.
invoice_date	The day when a transaction was generated.
shopping_mall	String variable of the name of the shopping mall where the transaction was made.

5. Project Deliverables

The project deliverables include the following:

- **Insights Narrative:** An analysis report describing the insights (PDF, DOC or TXT format).
- **Charts:** Screenshots of the charts created during the analysis.
- **Presentation:** A PowerPoint (PPT) presentation.